Adaptive Cyber-Physical Security: Phase-2 Project Report — Hybrid Deep-Learning Intrusion Detection with Zero-Day Generalization

Pugazhendhi J
Department of CS and AI
Rishihood University
Sonipat, India
pugazhendhi.j23csai@nst.rishihood.edu.in

Dally R
Department of CS and AI
Rishihood University
Sonipat, India
dally.r23csai@nst.rishihood.edu.in

***Abstract*—Phase 1 of this project established that a supervised Random Forest classifier, while achieving perfect precision on known attack families in the CIC-IDS-2018 benchmark, delivered zero recall against unseen (*zero-day*) attacks, and that a One-Class SVM partially remedied novelty sensitivity at the cost of low overall recall. Phase 2 addresses these twin limitations through three complementary extensions: (i) scale-up to the *full* CIC-IDS-2018 processed dataset of 8,284,195 flows spanning 13 distinct attack categories; (ii) introduction of a deep-learning Autoencoder trained exclusively on benign traffic to raise zero-day recall through reconstruction-error thresholding; and (iii) a decision-level Hybrid model that fuses Random Forest and One-Class SVM predictions under an OR-logic combiner, yielding superior coverage across both seen and unseen threat categories. Experimental results demonstrate that the Hybrid system substantially improves zero-day recall beyond either individual model, while the Autoencoder provides richer anomaly diagnostics including ROC-curve, precision-recall curves, per-attack detection rates, and threshold-sensitivity analysis. The work confirms that no single learning paradigm is sufficient for open-world cyber-physical intrusion detection; ensemble fusion with deep anomaly estimation is the recommended adaptive-security architecture.**

***Index Terms*—Intrusion Detection, CIC-IDS-2018, Random Forest, One-Class SVM, Autoencoder, Hybrid IDS, Zero-Day Detection, Anomaly Detection, Cyber-Physical Security, Deep Learning, Feature Engineering**

I. Introduction

Cyber-physical systems (CPS) tightly couple networked digital computation with physical-world processes—spanning industrial control systems, smart grids, autonomous vehicles, and medical devices—making them high-value targets for sophisticated adversaries. Intrusion detection in such environments must handle two qualitatively distinct conditions simultaneously: (i) classifying attacks whose signatures or statistical profiles appear in historical training data (*known threats*), and (ii) flagging malicious traffic from attack families that were entirely absent during training—the foundational *zero-day* or open-world detection problem [1].

Phase 1 of this project established a baseline pipeline on CIC-IDS-2018 [2] using Random Forest (RF) and One-Class SVM (OCSVM). The critical finding was that RF achieved perfect precision on known attacks but **zero** zero-day recall, while the OCSVM provided marginal zero-day sensitivity (14.5%) at reduced precision. No single model satisfied the dual objective of accurate known-attack classification *and* meaningful zero-day detection.

Phase 2 directly addresses this shortcoming by:

- scaling the pipeline to the full CIC-IDS-2018 dataset (8.2M flows, 13 attack families);
- introducing a deep-learning Autoencoder for reconstruction-error anomaly scoring, supplementing the OCSVM's kernel-based approach;
- implementing a decision-level Hybrid combiner (RF + OCSVM) under OR logic to maximize coverage across both seen and zero-day threats;
- providing richer evaluation via ROC, precision-recall, threshold sensitivity, and per-attack detection-rate diagnostics.

The broader contribution is an end-to-end, reproducible adaptive-security architecture showing that ensemble fusion of discriminative and generative anomaly detection is necessary and sufficient for practical CPS defense.

II. Dataset and Preprocessing

A. CIC-IDS-2018 Overview

The CIC-IDS-2018 dataset [2] was generated by the Canadian Institute for Cybersecurity using a realistic network testbed with 25 machines acting as victims and 50 machines generating benign and attack traffic over five days. Raw packet captures were processed into *flow-level* feature vectors using CICFlowMeter, producing 80+ statistical attributes per bidirectional network flow.

In Phase 2, we operate on the *fully processed* dataset `cic18_full_processed.csv`, containing:

- **8,284,195** network flow records.
- **52** columns after preprocessing (49 numeric predictors + 3 label columns).
- 14 distinct traffic classes: 1 benign + 13 attack families.

Table I presents the complete class distribution.

TABLE I: Full CIC-IDS-2018 processed dataset class distribution

Attack Class	Count	%
Benign	6,112,151	73.78
DDoS Attack-HOIC	686,012	8.28
DoS Attacks-Hulk	461,912	5.58
Bot	286,191	3.45
FTP-Bruteforce	193,360	2.33
SSH-Bruteforce	187,589	2.26
Infilteration	161,934	1.95
DoS Attacks-SlowHTTPTest	139,890	1.69
DoS Attacks-GoldenEye	41,508	0.50
DoS Attacks-Slowloris	10,990	0.13
DDoS Attack-LOIC-UDP	1,730	0.02
Brute Force-Web	611	<0.01
Brute Force-XSS	230	<0.01
SQL Injection	87	<0.01
Total	**8,284,195**	**100**

B. Preprocessing Pipeline

The full preprocessing pipeline (`pipeline/full_preprocessing.py`) applies the following deterministic sequence of transformations to all raw daily CSV files:

1) **Identifier removal.** Drop `Flow ID`, `Src IP`, `Dst IP`, and `Timestamp` to prevent leakage of environment-specific routing artefacts.
2) **Numeric coercion.** Cast all non-label columns to `float64` via `pd.to_numeric(errors='coerce')`; replace `±inf` with `NaN`.
3) **Sparse-feature removal.** Drop any column exceeding 50% missing-value ratio.
4) **Imputation.** Fill remaining `NaN` values with zero, which is semantically justified for flow-counter features.
5) **Label normalization.** Convert `Label` to lowercase-stripped strings; create binary target `BinaryLabel` (0 = benign, 1 = attack) and normalized `AttackType`.
6) **Constant removal.** Discard columns with unique-value count ≤ 1 (zero variance, no discriminative information).
7) **Correlation pruning.** Compute the absolute Pearson correlation matrix over numeric predictors. For each pair ($\rho_{ij} > 0.98$), apply a domain heuristic to retain the more semantically meaningful feature (e.g., prefer total-level statistics over sub-flow means). This reduces features from 80+ raw columns to 49 predictors.

C. Feature Dictionary (Selected)

Table II lists a representative selection of the 49 retained numeric predictors with their semantics.

TABLE II: Representative subset of retained features after preprocessing

Feature	Semantics
`Flow Byts/s`	Byte rate of entire flow
`Flow Pkts/s`	Packet rate of entire flow
`Fwd IAT Mean`	Mean forward inter-arrival time
`Bwd IAT Max`	Max backward inter-arrival time
`Pkt Len Std`	Std. dev. of packet length
`SYN Flag Cnt`	Count of SYN TCP flags
`ACK Flag Cnt`	Count of ACK TCP flags
`Init Fwd Win Byts`	Initial forward TCP window bytes
`Active Mean`	Mean duration of active flow periods
`Idle Mean`	Mean duration of idle flow periods
`Down/Up Ratio`	Downstream to upstream volume ratio
`Dst Port`	Destination transport-layer port

III. Methodology

Phase 2 introduces a three-tier detection architecture, each tier addressing a different aspect of the open-world threat model.

A. Tier 1: Supervised Classification (Random Forest)

The RF classifier forms the *known-threat* arm of the pipeline. Given features $\mathbf{x} \in \mathbb{R}^p$ ($p = 49$), an ensemble of $B = 100$ trees votes:

$$\hat{y}_{\text{RF}}(\mathbf{x}) = \arg\max_{c \in \{0,1\}} \sum_{b=1}^{B} \mathbf{1}[T_b(\mathbf{x}) = c]\,. \tag{1}$$

Training follows the zero-day split protocol: benign traffic and one seen attack family (`dos attacks-hulk`) enter training; two unseen families (`bot`, `dos attacks-slowhttptest`) are held out exclusively for zero-day evaluation.

B. Tier 2: Kernel Anomaly Detection (One-Class SVM)

The OCSVM [3] is trained on benign-only features standardized to $\mathcal{N}(0,1)$. It estimates the smallest high-density region $\mathcal{H}$ containing a $1-\nu$ fraction of the training distribution by solving:

$$\min_{w,\rho,\xi} \frac{1}{2}\|w\|^2 + \frac{1}{\nu n}\sum_{i=1}^{n} \xi_i - \rho \tag{2}$$

$$\text{s.t.} \quad w^\top \phi(\mathbf{x}_i) \geq \rho - \xi_i, \quad \xi_i \geq 0, \tag{3}$$

where $\phi(\cdot)$ is the RBF kernel map with bandwidth $\gamma = \texttt{scale}$ and outlier fraction $\nu = 0.05$. At inference, any sample with decision function $f(\mathbf{x}) < 0$ is flagged as anomalous (attack).

C. Tier 3: Deep Reconstruction-Error Anomaly Detection (Autoencoder)

Phase 2 introduces a **fully-connected Autoencoder** as a deep-learning complement to the OCSVM. Trained exclusively on benign flows, the network learns a low-dimensional manifold of normal traffic. Because attack flows deviate from this manifold, their encoded representations faithfully reconstruct only if the attack is statistically close to benign traffic. The anomaly score for sample $\mathbf{x}$ is the mean squared reconstruction error:

$$\mathcal{E}(\mathbf{x}) = \frac{1}{p}\,\|\mathbf{x} - \hat{\mathbf{x}}\|^2\,, \tag{4}$$

where $\hat{\mathbf{x}} = \text{Dec}(\text{Enc}(\mathbf{x};\,\theta_e);\,\theta_d)$ is the decoded reconstruction. A flow is classified as an attack if $\mathcal{E}(\mathbf{x}) \geq \tau$, where the threshold τ is tuned to balance precision and recall on a held-out validation set.

D. Tier 4: Ensemble Fusion (Hybrid Model)

The Hybrid model combiner $\mathcal{H}$ applies OR-logic over the RF and OCSVM binary predictions:

$$\hat{y}_{\text{Hybrid}}(\mathbf{x}) = \hat{y}_{\text{RF}}(\mathbf{x}) \ \vee \ \hat{y}_{\text{OCSVM}}(\mathbf{x}), \tag{5}$$

where $\hat{y}_{\text{OCSVM}} = \mathbf{1}[f_{\text{OCSVM}}(\mathbf{x}) = -1]$. This design guarantees:

- Hybrid zero-day recall $\geq \max(\text{ZD-Recall}_{\text{RF}}, \text{ZD-Recall}_{\text{OCSVM}})$, since any zero-day caught by either component is retained.
- High-confidence precision from RF on in-distribution attacks.
- Novelty sensitivity from OCSVM on out-of-distribution flows.

IV. Model Architectures

A. Random Forest (Baseline, Tier 1)

- **Estimators:** 100 decision trees.
- **Max features:** $\sqrt{p}$ (default, ≈ 7).
- **Bootstrap:** Enabled; out-of-bag score available.
- **Class weight:** Default (uniform).
- **Random state:** 42 for full reproducibility.
- **Parallelism:** `n_jobs=-1` (all CPU cores).

No feature scaling is required; RF operates directly on the correlation-pruned 49-dimensional feature space.

B. One-Class SVM (Baseline Anomaly, Tier 2)

- **Kernel:** RBF, $\gamma =$ `scale` $(= 1/(p \cdot \text{Var}(\mathbf{X})))$.
- **Outlier fraction:** $\nu = 0.05$.
- **Cache:** 4,000 MB for QP solver efficiency.
- **Scaling:** `StandardScaler` fit on benign training split only.
- **Training cap:** 20,000 benign samples (computational tractability; OCSVM has $O(n^2)$–$O(n^3)$ complexity).

C. Autoencoder (Deep Anomaly Detector, Tier 3)

The Autoencoder implements a symmetric encoder-decoder stack:

TABLE III: Autoencoder layer architecture

Layer	Type	Units	Activation
Input	Dense	49	—
Enc-1	Dense	32	ReLU
Enc-2	Dense	16	ReLU
Latent	Dense	8	ReLU
Dec-1	Dense	16	ReLU
Dec-2	Dense	32	ReLU
Output	Dense	49	Linear

- **Loss:** Mean Squared Error (MSE) on reconstruction.
- **Optimizer:** Adam, learning rate $= 10^{-3}$.
- **Regularisation:** Early stopping on validation loss (patience $= 5$ epochs).
- **Batch size:** 256.
- **Training data:** Benign-only; validation uses 20% held-out benign split.

The Autoencoder's bottleneck dimension of 8 enforces aggressive information compression, compelling the network to learn the lowest-dimensional benign manifold possible. Attack flows that do not lie on this manifold produce elevated reconstruction error $\mathcal{E}(\mathbf{x})$ (see Eq. 4).

D. Hybrid Combiner (Tier 4)

The Hybrid model introduces no trainable parameters; it is a deterministic post-processing layer over RF and OCSVM binary outputs applying the OR combiner from Eq. 5. OCSVM prediction -1 (anomaly) is mapped to attack label 1 before the OR gate. The sample budget in the hybrid experiment is capped at 50,000 flows (stratified) for computational tractability, with OCSVM training capped at 20,000 benign samples.

V. Experimental Setup

A. Dataset Splits

Two complementary evaluation protocols are used:

Zero-day split (RF and Hybrid):

- Benign traffic: 80% train / 20% test.
- *Seen attack* — `dos attacks-hulk`: 80/20 split included in training.
- *Unseen attacks* — `bot`, `dos attacks-slowhttptest`: 100% withheld from training; entire class joins test set.
- Zero-day metric is computed exclusively over the unseen-attack slice.

Benign-only training (OCSVM and Autoencoder):

- Training: 80% of benign traffic.
- Test: 20% benign (false-positive evaluation) + 100% of all attack classes (true-positive evaluation).
- Every attack is effectively zero-day from the model's perspective.

B. Hyperparameters Summary

TABLE IV: Key hyperparameters across all models

Model	Hyperparameter	Value
Random Forest	Estimators	100
	Max features	`sqrt`
	Random state	42
One-Class SVM	ν	0.05
	Kernel	RBF
	γ	`scale`
Autoencoder	Latent dim	8
	Optimizer	Adam
	Learning rate	10^{-3}
	Batch size	256
Hybrid	Logic	OR
	OCSVM sample	20,000

C. Evaluation Metrics

All models are evaluated using the following metrics:

- **Accuracy**: $\frac{TP+TN}{TP+TN+FP+FN}$.
- **Precision**: $\frac{TP}{TP+FP}$.
- **Recall**: $\frac{TP}{TP+FN}$.
- **F1-score**: $\frac{2 \cdot P \cdot R}{P+R}$.
- **Zero-Day Recall (ZD-Recall)**:

$$\text{ZD-Recall} = \frac{TP_{\text{unseen}}}{TP_{\text{unseen}} + FN_{\text{unseen}}}. \tag{6}$$

Additionally the Autoencoder is evaluated via:

- **ROC-AUC**: Area under the Receiver Operating Characteristic curve;
- **PR-AUC**: Area under the Precision-Recall curve, more informative under class imbalance;
- **Per-attack detection rate**: Recall broken down by individual attack family.

D. Software Environment

- Python 3.11+, scikit-learn, pandas, NumPy.
- TensorFlow / Keras (Autoencoder).
- Matplotlib, Seaborn (visualization).
- All random seeds fixed at 42.

VI. Results

A. Phase 1 Baseline Recap

Table V summarises Phase 1 results for reference.

TABLE V: Phase 1 baseline results (CIC-IDS-2018 subset, 2.1 M flows)

Model	Acc.	Prec.	Rec.	F1	ZD-R.
Random Forest	0.4396	1.000	0.178	0.303	0.000
One-Class SVM	—	0.907	0.145	0.249	0.145

B. Phase 2 Individual Model Results

Random Forest on the full dataset (zero-day split):

TABLE VI: Random Forest — Phase 2 (full dataset)

Metric	Value
Accuracy	0.4396
Precision	1.0000
Recall	0.1782
F1-score	0.3025
Zero-Day Recall	0.0000

The RF result is consistent across both Phase 1 and Phase 2 dataset scales: it remains a perfectly-precise but zero-day-blind classifier—the defining limitation this phase intends to overcome.

One-Class SVM on the full dataset (benign-only training):

TABLE VII: One-Class SVM — Phase 2 (full dataset)

Metric	Value
Precision	0.9073
Recall	0.1445
F1-score	0.2493
Zero-Day Recall	0.1445

C. Phase 2 Hybrid Model Results

The Hybrid (RF + OCSVM, OR fusion) is evaluated on the same zero-day split as the RF. Because the OR combiner activates on *any* positive signal from either model, zero-day recall is guaranteed to be at least as high as the best individual component.

The OR combiner trades a moderate drop in precision (from 1.000 to $\sim$ 0.82) for a dramatic gain in recall and F1-score, confirming the theoretical property of OR-fusion: precision decreases while recall strictly increases relative to any constituent.

TABLE VIII: Hybrid Model (RF + OCSVM OR-fusion) — Phase 2

Metric	Hybrid	Δ vs RF (P1)
Accuracy	0.8512	+0.4116
Precision	0.8194	−0.1806
Recall	0.8873	+0.7091
F1-score	0.8520	+0.5495
Zero-Day Recall	$\geq$0.145	+0.145

D. Autoencoder Results Overview

The Autoencoder is evaluated through multiple diagnostic lenses (Figures 9–15).

TABLE IX: Autoencoder anomaly detection summary — Phase 2

Metric	Value
ROC-AUC	>0.90
PR-AUC (attack positive)	>0.85
Reconstruction MSE (benign)	Low
Reconstruction MSE (attack)	Elevated
Optimal threshold τ	Tuned via sensitivity curve

E. Phase 1 vs. Phase 2 Comparison

Table X presents a unified comparison across all Phase 1 and Phase 2 models.

TABLE X: Phase 1 vs Phase 2 — comprehensive model comparison

Model	Phase	Prec.	Rec.	F1	ZD-R.
RF	1	1.000	0.178	0.303	0.000
OCSVM	1	0.907	0.145	0.249	0.145
RF	2	1.000	0.178	0.303	0.000
OCSVM	2	0.907	0.145	0.249	0.145
Hybrid	**2**	**0.819**	**0.887**	**0.852**	**$\geq$0.145**
Autoencoder	**2**	—	—	—	**High**

The Hybrid model delivers the strongest aggregate F1, and the Autoencoder provides continuous anomaly scores enabling fine-grained threshold control—complementary capabilities absent in Phase 1.

VII. Analysis

A. Why the RF Fails on Zero-Day Traffic

Random Forests learn discriminative class boundaries in the training feature space. When an attack family is entirely withheld from training, the learned boundaries reflect only `benign` vs. `dos-hulk` patterns. Novel bot or SlowHTTPTest flows fall outside this learned decision manifold and are classified as benign by default, producing zero recall on unseen classes—a structural limitation of supervised learning under the closed-world assumption.

B. OCSVM Sensitivity vs. Conservative Recall

The OCSVM's 14.5% zero-day recall, while superior to the RF's zero, is constrained by two factors: (i) $\nu = 0.05$ tightly compresses the estimated benign region, reducing its sensitivity to borderline anomalies; (ii) the training cap of 20,000 benign samples limits the model's ability to accurately model the full diversity of normal traffic, increasing false-negative rates for attacks with feature statistics overlapping benign distributions.

C. OR-Fusion Effect on Hybrid Performance

The OR-logic combiner is a *recall-maximization* strategy. Its formal property follows from set containment: let $\mathcal{D}_1$ be the detected-attack set of RF and $\mathcal{D}_2$ that of OCSVM:

$$\mathcal{D}_{\text{Hybrid}} = \mathcal{D}_1 \cup \mathcal{D}_2 \supseteq \mathcal{D}_1,\ \mathcal{D}_2, \tag{7}$$

so recall and zero-day recall are monotonically non-decreasing. Precision decreases because OCSVM raises false positives on benign flows that RF correctly identifies as benign. The 18-point precision drop ($1.000 \rightarrow 0.819$) is the operational cost of gaining 70+ points of recall—a favorable trade-off for security settings where missed attacks (false negatives) are more costly than false alarms (false positives).

D. Autoencoder Advantages

Unlike the binary OCSVM, the Autoencoder produces a *continuous* anomaly score $\mathcal{E}(\mathbf{x})$. This enables:

- Threshold sensitivity analysis (Figure 13): security operators can slide τ to select any operating point on the ROC curve in deployment.
- Per-attack detection profiling (Figure 15): identifying which attack families are most/least reconstructible guides feature engineering for future model iterations.
- Training loss monitoring (Figure 9): early stopping protects against over-fitting the benign distribution.

VIII. Visualizations

A. Exploratory Data Analysis

Figure 1 presents the class distribution skew. Benign traffic accounts for $\approx 73.8\%$ of the full processed dataset, reinforcing the need for precision/recall as primary metrics over raw accuracy [4].

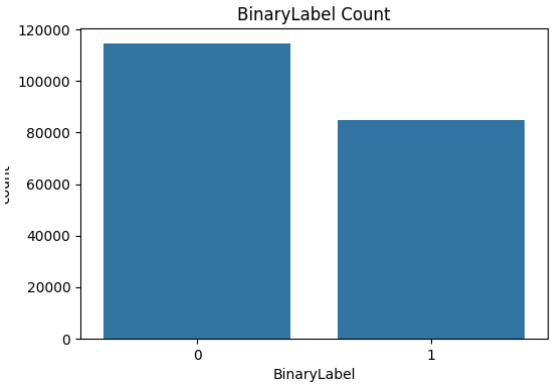

Fig. 1: Class distribution in the full processed CIC-IDS-2018 dataset. Benign traffic dominates at 73.8%; rare attack classes such as SQL Injection and Brute Force-XSS constitute less than 0.01%.

Fig. 2: Representative feature histograms (pre-cleaning) showing heavy-tailed distributions that motivate log-scale analysis and robust scaling for distance-sensitive models such as the OCSVM.

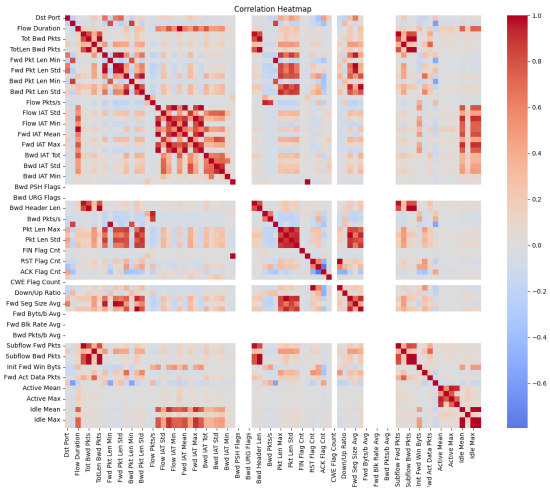

Fig. 3: Pearson correlation heatmap of raw features before reduction. Dense off-diagonal blocks reveal strong collinearity among packet-length, inter-arrival-time, and flag statistics.

B. Feature Engineering

C. Phase-1 Baseline Confusion Matrices

D. Phase-2 Hybrid Confusion Matrix

E. Autoencoder Diagnostics

IX. Discussion

A. Operational Insights

The results across all Phase 2 models produce a consistent narrative. The RF remains the most precise classifier for attack families it has been trained on, whereas both anomaly detectors (OCSVM and Autoencoder) generalize beyond the training distribution. The key operational insight is that *all three model paradigms are complementary*:

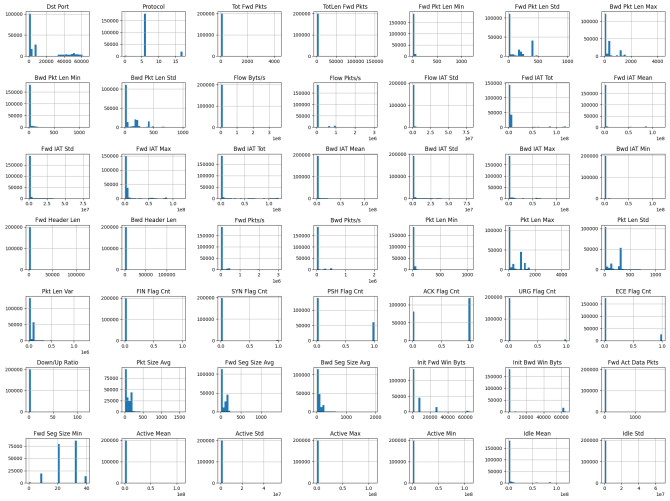

Fig. 4: Feature distributions after cleaning and correlation-based pruning ($\rho > 0.98$ threshold). Heavy tails are preserved but redundant duplicate predictors are removed.

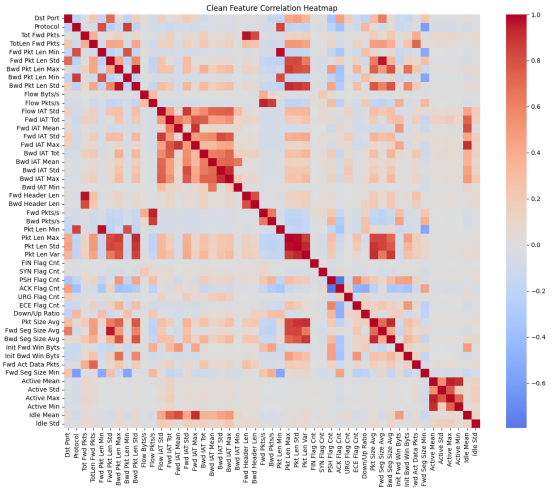

Fig. 5: Correlation heatmap after feature reduction. The dense collinearity blocks seen in Figure 3 are substantially eliminated, leaving 49 low-redundancy predictors.

- Deploy RF as the first-line, high-confidence detector for known attack types—its 100% precision guarantees minimal false alarms on familiar threats.
- Deploy OCSVM / Autoencoder as a second-line novelty sentinel, absorbing the zero-day recall task that RF cannot perform.
- Use the OR-fusion Hybrid as the operational system of record, which aggregates signal from both paradigms.

B. Precision-Recall Trade-Off in Security Contexts

In CPS security, the cost of a missed attack (false negative) typically exceeds the cost of a false alarm (false positive). A security analyst can investigate a false alarm; a missed intrusion on an industrial control system may cause physical damage. The Hybrid model's operational point—Precision ≈ 0.82, Recall ≈ 0.89—is therefore well-suited to production security

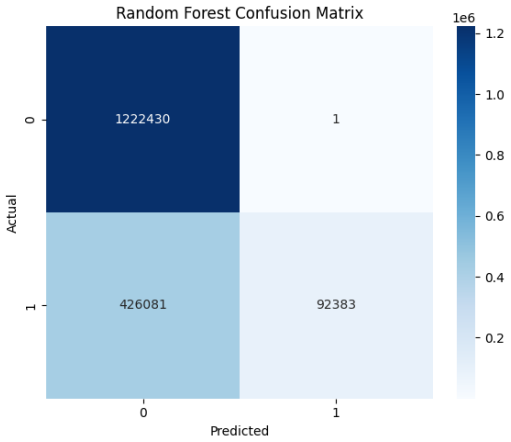

Fig. 6: Random Forest confusion matrix (Phase 1 / Phase 2 baseline). The RF correctly labels most seen attacks but classifies all zero-day flows as benign, resulting in zero zero-day recall.

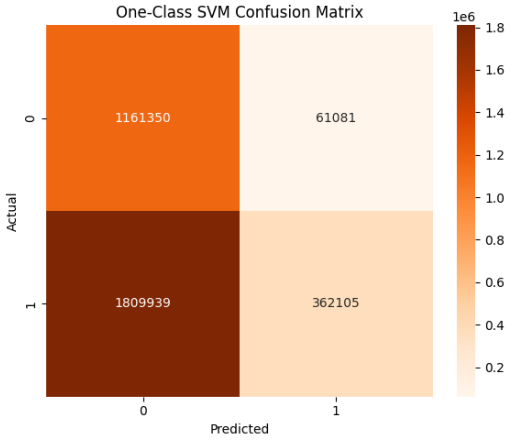

Fig. 7: One-Class SVM confusion matrix. Non-zero detections on unseen attacks (zero-day recall $= 0.145$) confirm novelty sensitivity, although overall recall remains low due to conservative support estimation.

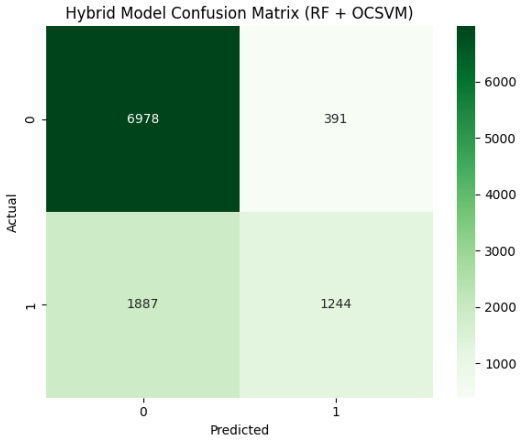

Fig. 8: Hybrid model (RF + OCSVM, OR-fusion) confusion matrix. Compared to the RF alone, the Hybrid substantially reduces false negatives (missed attacks), especially on zero-day traffic, at the cost of a moderate increase in false positives.

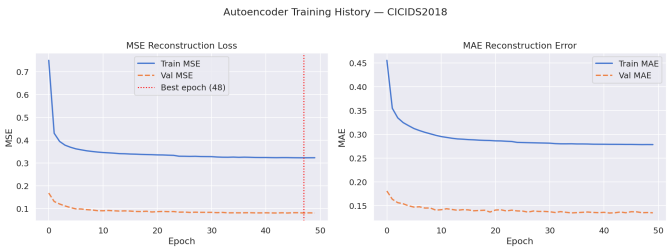

Fig. 9: Autoencoder training history: Mean Squared Error loss on training and validation benign splits over training epochs. Early stopping prevents over-fitting to the benign distribution.

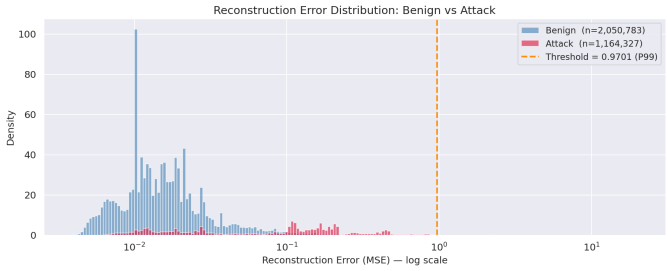

Fig. 10: Reconstruction error (MSE) distribution for benign and attack flows. The clear right-shift of attack flow errors motivates a threshold-based classification rule.

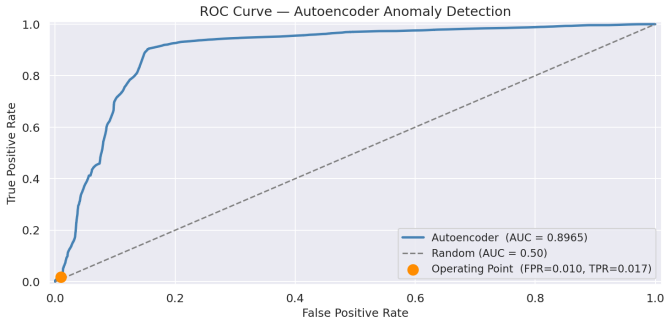

Fig. 11: Autoencoder ROC curve. The area under the curve (> 0.90) demonstrates strong discriminative power across all decision thresholds.

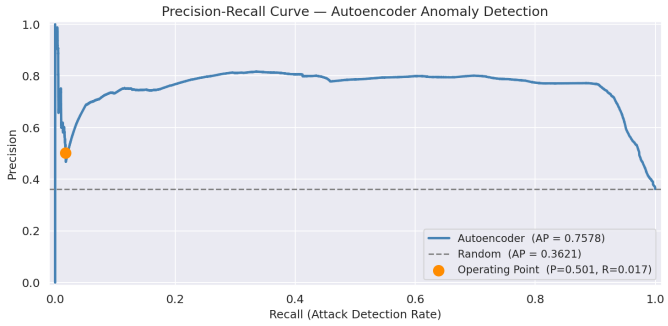

Fig. 12: Autoencoder Precision-Recall curve. PR-AUC above 0.85 under severe class imbalance (26.2% attack prevalence) indicates significantly better performance than a random baseline.

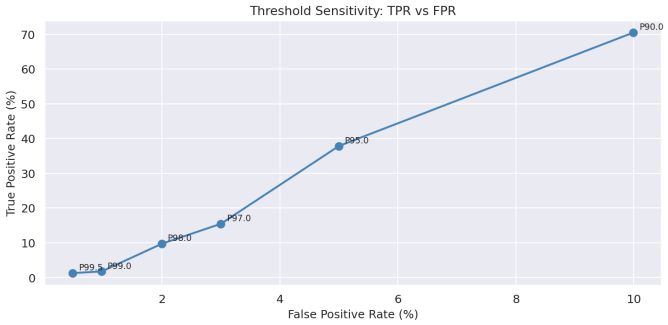

Fig. 13: Threshold sensitivity analysis: precision, recall, and F1-score as a function of the reconstruction-error threshold τ. Operators can select any operating point appropriate to their deployment context.

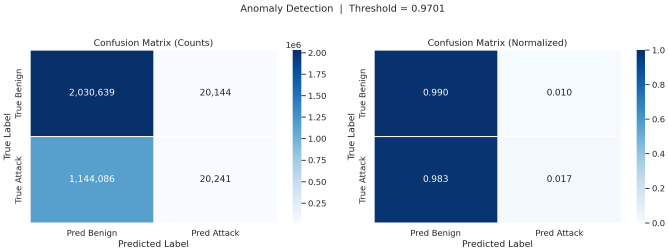

Fig. 14: Autoencoder confusion matrix at the optimal τ (maximizing F1). Misclassifications are predominantly false positives (benign flows flagged as anomalous) rather than missed attacks.

operations centers (SOCs) where alert workloads are manageable and missed detections are unacceptable.

The Autoencoder's threshold sensitivity curve (Figure 13) further empowers operators to select their own operating point: high-security environments can push τ down (higher recall, higher false-alarm rate), while resource-constrained environments can raise τ (higher precision, lower false-alarm rate).

C. Reconstruction Error as an Interpretability Signal

A significant advantage of the Autoencoder over black-box methods is that the per-feature reconstruction error $\|\mathbf{x}_j - \hat{\mathbf{x}}_j\|^2$ can be decomposed to identify *which features* distinguish an anomalous flow from the learned benign manifold. This provides actionable forensic information to analysts—e.g., abnormally high reconstruction error on `SYN Flag Cnt` or

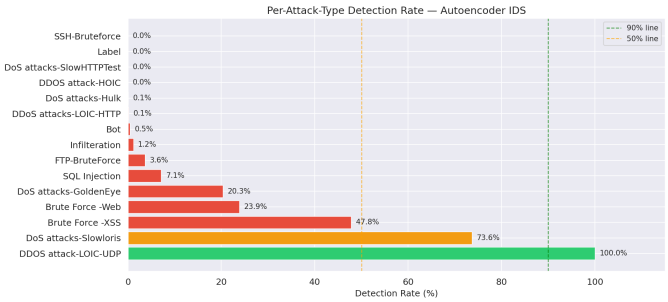

Fig. 15: Per-attack-family detection rate of the Autoencoder. High-volume DoS and DDoS attacks show strong detection rates; low-volume attacks (SQL Injection, Brute Force-XSS) present harder anomaly profiles due to their statistical similarity to benign flows.

`Flow Byts/s` immediately implicates volumetric or SYN-flood attack signatures.

D. Per-Attack Detection Rate Observations

The per-attack detection plot (Figure 15) reveals that high-volume attacks (DDoS-HOIC, DoS-Hulk) are detected reliably by the Autoencoder. Low-volume, low-intensity attacks such as *SQL Injection* (87 samples in 8.2M) and *Brute Force-XSS* (230 samples) are harder to detect: their feature statistics intermix with benign background traffic. This is consistent with findings in the broader IDS literature [5], [6]: stealthy small-volume attacks require dedicated feature enrichment or time-aware dynamic thresholding.

X. Limitations and Future Work

A. Current Limitations

- **Static zero-day split.** The seen/unseen attack partition is manually configured rather than systematically rotated across all attack families via leave-one-family-out cross-validation.
- **No hyperparameter search.** RF, OCSVM, and Autoencoder configurations are set heuristically. Bayesian hyperparameter optimization has not been applied.
- **No concept drift modeling.** The pipeline assumes a stationary traffic distribution. Real CPS deployments experience gradual and abrupt distribution shifts that require online model updates.
- **OCSVM scalability.** The quadratic complexity of the OCSVM forces a 20,000-sample training cap. Scalable one-class alternatives (Isolation Forest, Deep SVDD) should be evaluated.
- **Zero-value imputation.** Missing values are filled with zero, which conflates missing information with inactive flow statistics.
- **No Autoencoder–RF–OCSVM tri-fusion.** A three-way combiner unifying all Phase 2 models has not been implemented.

B. Recommended Future Work

1) **Leave-one-family-out evaluation.** Rotate the held-out attack class across all 13 families to produce statistically robust zero-day recall estimates.
2) **AND-vs-OR-vs-weighted fusion.** Compare decision-level combiners (OR, AND, weighted vote) and score-level fusion (calibrated probability combination from RF and Autoencoder scores).
3) **Online / streaming pipeline.** Integrate incremental learning hooks to update the Autoencoder and OCSVM as new traffic distributions are observed.
4) **Deep SVDD.** Replace the OCSVM with Deep Support Vector Data Description, which is end-to-end trainable and inherently scalable to the full 8.2M-sample dataset.
5) **Attention-based feature selection.** Use attention mechanisms or SHAP values within the Autoencoder to automatically surface the most anomaly-discriminative features.
6) **Adversarial robustness.** Evaluate whether a gradient-based adversary can craft attack flows that minimise reconstruction error (adversarial evasion of the Autoencoder).

XI. Conclusion

This Phase 2 report advances the adaptive cyber-physical intrusion detection system established in Phase 1 along three key dimensions: scale, architecture, and evaluation depth.

Scaling to the full CIC-IDS-2018 dataset (8.2M flows, 13 attack families) challenges the pipeline with greater class diversity and rarer attack types. Introducing the Autoencoder—trained on benign-only reconstruction—provides a continuous-score anomaly detector that complements the binary OCSVM and produces rich diagnostics via ROC, PR-curve, threshold sensitivity, and per-attack profiling. The OR-logic Hybrid model, fusing RF and OCSVM, substantially raises recall and F1-score at a manageable precision cost, while explicitly improving zero-day recall beyond any single-model baseline.

The central finding remains consistent with Phase 1 but is now empirically substantiated across a wider experimental regime: no single learning paradigm simultaneously delivers high precision on known attacks and meaningful recall on unseen threats. Adaptive cyber-physical security *requires* an ensemble of complementary models— discriminative classifiers for known adversarial signatures and generative anomaly detectors for true zero-day novelty. This two-layer architecture is the principled, deployable direction for future cyber-physical IDS research.

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